Computer Science & IT



Data Structure & Programming

Graph & Hashing

Lecture No. 02



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Recap of Previous Lecture



Topic — Graph Representation

Topic — Adjacency list

Topic — Adjacency Matrix

Topic — BFS traversal (DFS, BFS repeated)

Topic

Slide

Topics to be Covered



- Topic — Direct Addressing
- Topic — Hashing
- Topic — Collision
- Topic — chaining
- Topic — Linear probing

Slide



Topic : Hashing

Search

- Array (n elements)
  - ordered — Binary Search $O(\log_2 n)$
  - unordered — Linear Search $O(n)$
- Linked list
  - ordered — $O(n)$
  - unordered — $O(n)$
- Tree — $O(n)$
- BST
  - BST - $O(n)$
  - balanced BST — $O(\log_2 n)$
- AVL - Height balanced — $O(\log_2 n)$

Slide



Topic : Hashing

Direct Addressing

In direct Addressing A Large array is used and index of array represent the key value.

Key : 9824

A [9824] — data will Located

Search time complexity : O(1)

Slide



Topic : Hashing

Disadvantage of direct Addressing

1. Large array required phone No. 9211145367

2 Some Index may not represent the key value.

Space wastage

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Topic : Hashing

To overcome disadvantage a small size array called Hash table is used and to distribute the key a function called Hash function is used.

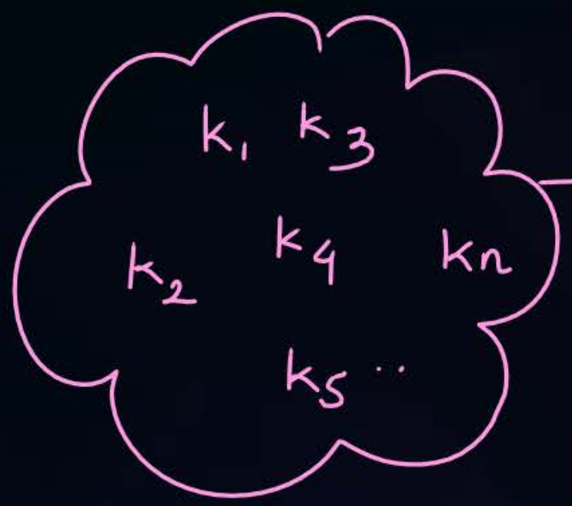
Direct Addressing key k stored in Index k

In Hashing key k stored in h(k)

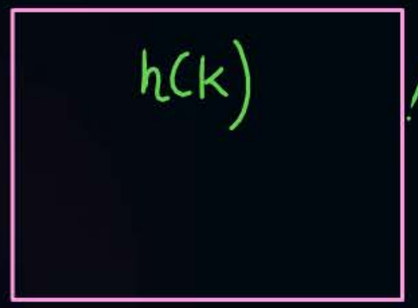
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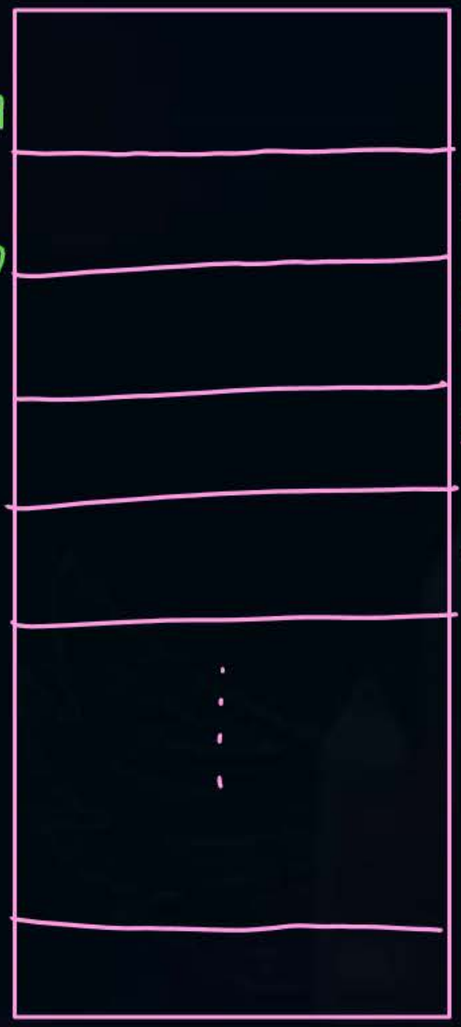


Topic : Hashing



hash function





hash table of Size m is used

Index start from 0

Slide



Topic : Hashing

Hash function The hash function should distribute the key in uniform manner

Each location of Hash table refered as slot, bucket

all location must have equal probability.

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Topic : Hashing

Which one of the following hash functions on integers will distribute keys most uniformly over 10 buckets numbered 0 to 9?

(A) $h(i) = i^2 \bmod 10$

(B) $h(i) = i^3 \bmod 10$

(C) $h(i) = (11 * i^2) \bmod 10$

(D) $h(i) = (12 * i^2) \bmod 10$

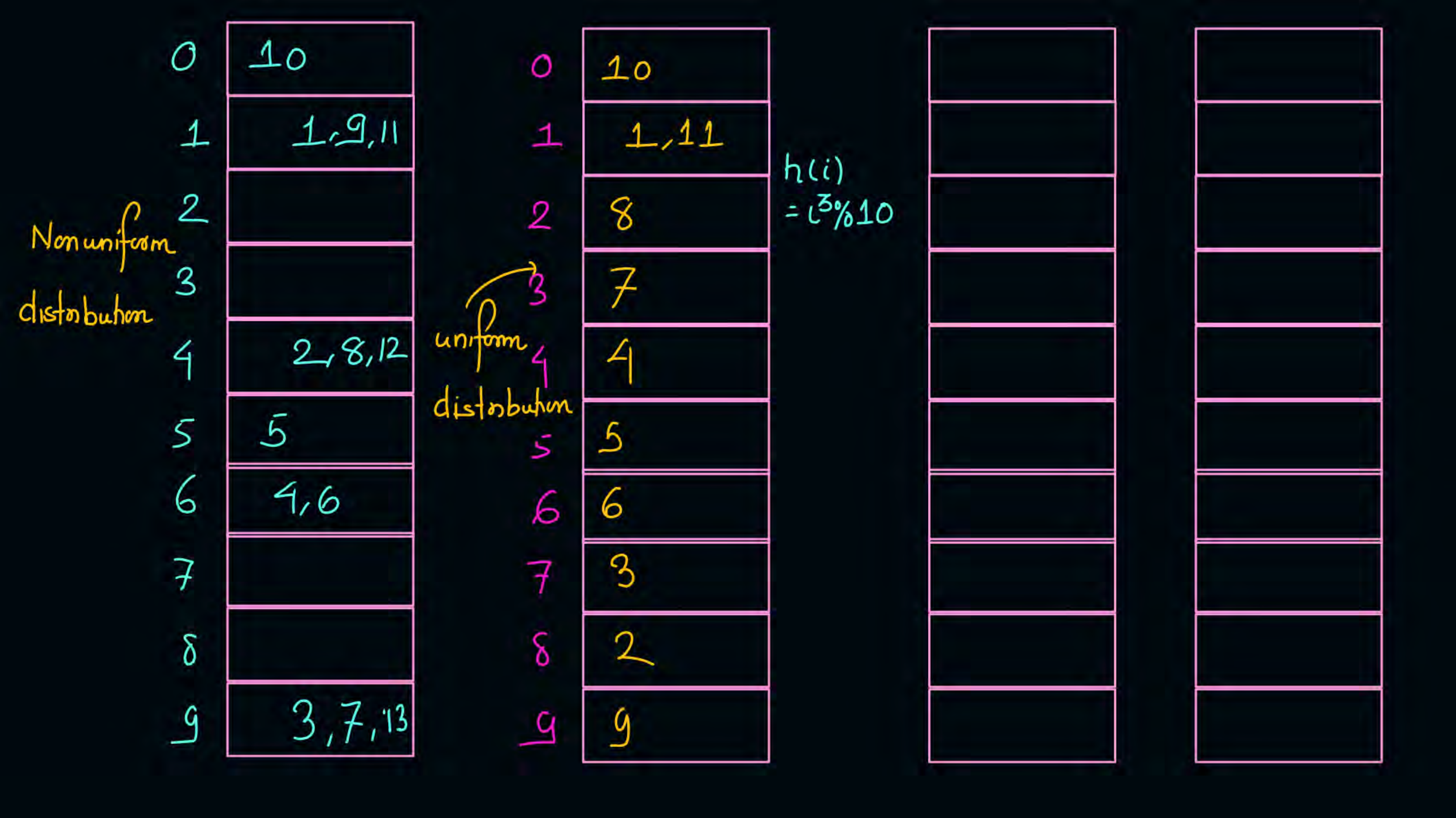
Non uniform

(A) $h(1) = 1^2 \% 10 = 1$

$h(2) = 2^2 \% 10 = 4$

$h(3) = 3^2 \% 10 = 9$

Slide


0
10
1
1,9,11
2
3
4
2,8,12
5
5
6
4,6
7
8
9
3,7,13
Non uniform distribution
0
10
1
1,11
2
8
3
7
4
4
5
5
6
6
7
3
8
2
9
9
uniform distribution
h(i) = i^3 % 10



Load factor :

Load factor described as

Ratio between No. of records (n) to No. of slots (m)

denoted by α

$$\alpha = \frac{n}{m}$$



Topic : Hashing

#Q Given a has table T with 25 slots that stores 2000 elements, the load factor α for T is 80.

$$\alpha = \frac{2000}{25} = 80$$

Slide



Topic : Hashing

collision : Two keys mapped to same location leads to collision

$h(k) = k^2 \bmod 10$

$h(4) = 4^2 \bmod 10 = 6$

$h(6) = 6^2 \bmod 10 = 6$

} Collision

Slide



Topic : Hashing

Collision Resolution Technique

open hashing → The collided key stors outside of Hash table. Dynamic and we can't predict No of keys in advance

Closed Hashing / Open Addressing → The collided key stored within the table in different Location.

Slide



Topic : Hashing

Open Hashing :- chaining

In chaining each time collision occurs the collided key stored in a linked list Node outside of table

Slide

Let us see the following example to get better idea. If we have some elements like $\{15, 47, 23, 34, 85, 97, 65, 89, 70\}$. And our hash function is $h(x) = x \bmod 7$.



No of collision = 4

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Topic : Hashing

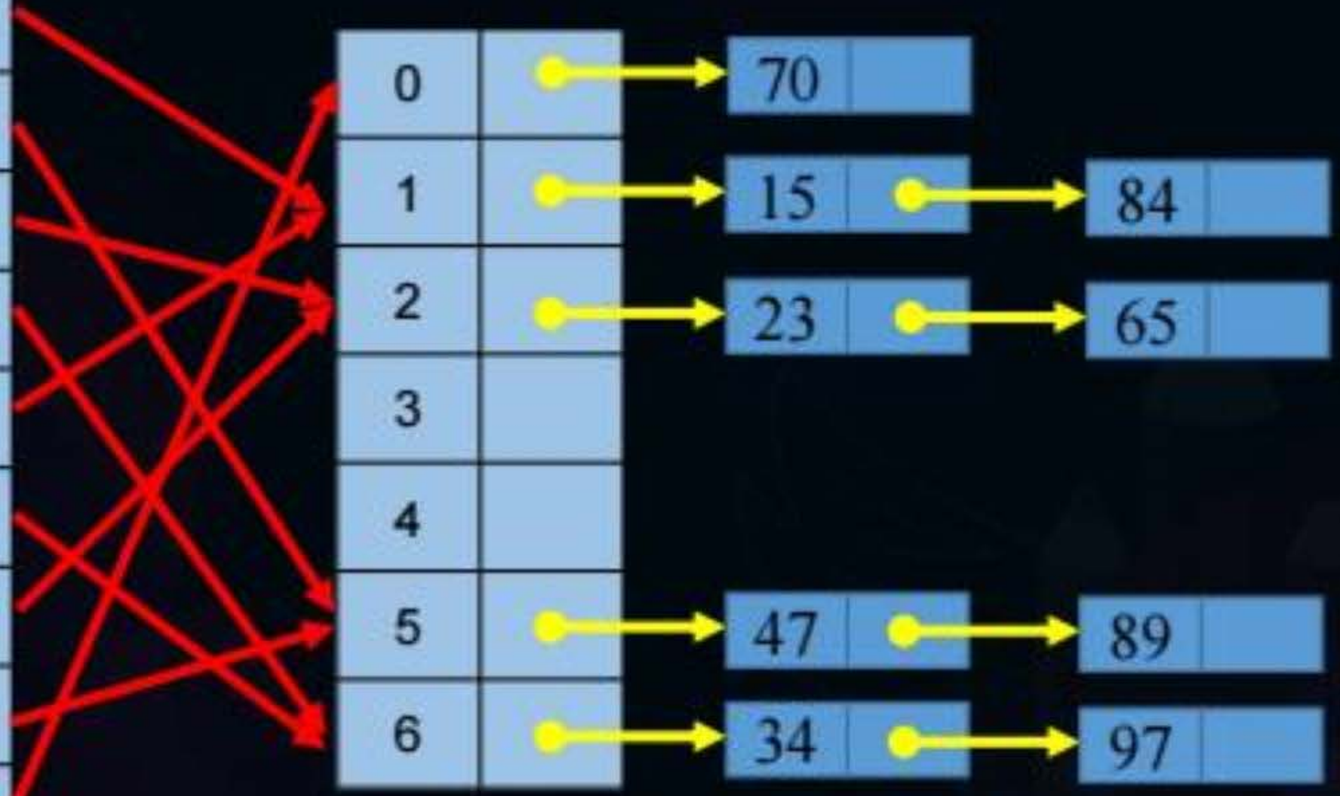
Example

Let us see the following example to get better idea. If we have some elements like $\{15, 47, 23, 34, 85, 97, 65, 89, 70\}$. And our hash function is $h(x) = x \bmod 7$.



Topic : Hashing

$Key - x$	$h(x) = x \bmod 7$
15	1
47	5
23	2
34	6
85	1
97	6
65	2
89	5
70	0



Slide



Topic : Hashing

#Q. Consider a hash table with 9 slots. The hash function is The collisions are resolved by chaining.

$h(x) = x \bmod 9$

The following 9 keys are inserted in the order: 5, 28, 19, 15, 20, 33, 12, 17, 10.

The maximum, minimum, and average chain lengths in the hash table, respectively, are

(A) 3, 0, and 1

(B) 3, 3, and 3

(C) 4, 0, and 1

(D) 3, 0, and 2

Slide

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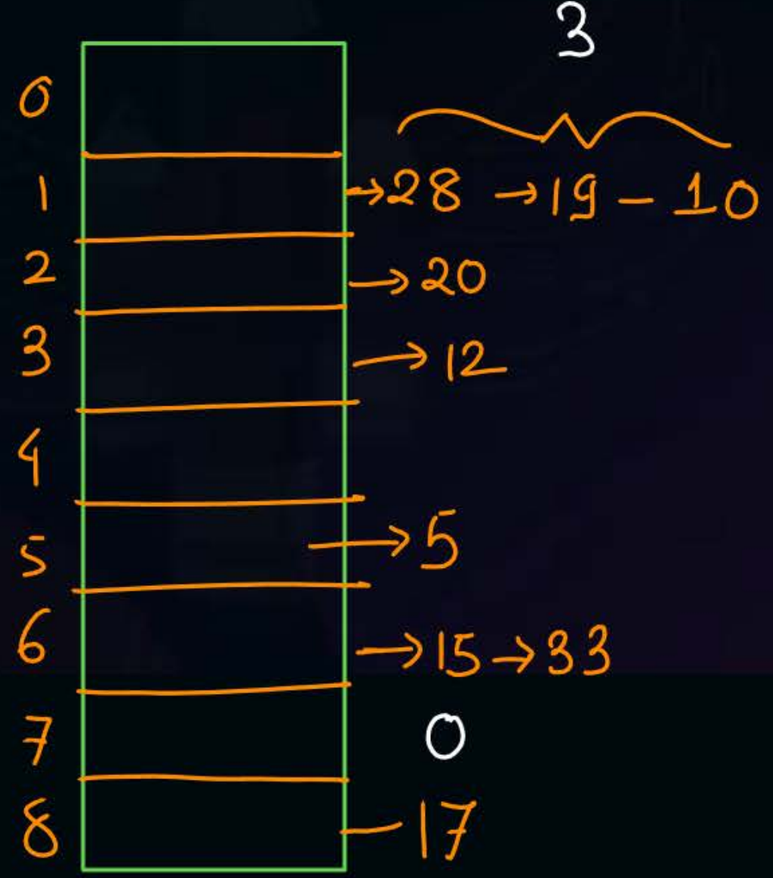
(A) 3, 0, and 1

(B) 3, 3, and 3

(C) 4, 0, and 1

(D) 3, 0, and 2

$h(x) = x \bmod 9$



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Topic : Hashing

Q. Consider a hash table with 100 slots. Collisions are resolved using chaining.

Assuming simple uniform hashing, what is the probability that the first 3 slots are unfilled after the first three insertions?

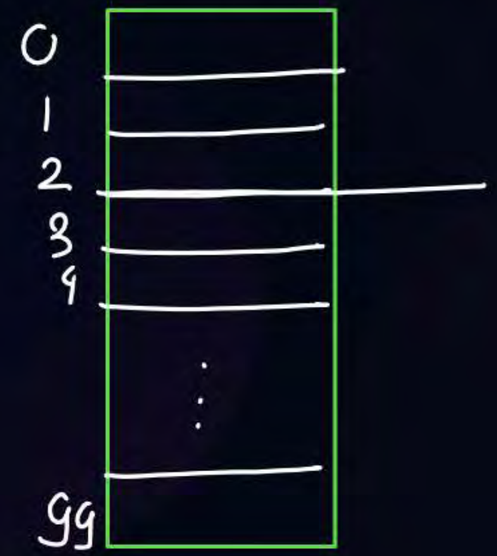
(A) $(97 \times 97 \times 97)/100^3$

(B) $(99 \times 98 \times 97)/100^3$

(C) $(97 \times 96 \times 95)/100^3$

(D) $(97 \times 96 \times 95)/(3! \times 100^3)$

probability formula : $\frac{\text{favorable}}{\text{total}}$



First Second Third

$\frac{97}{100} * \frac{97}{100} * \frac{97}{100}$

Slide



Topic : Hashing

Closed Hashing : Collided key stored within the table in different location

1. Linear probing
2. Quadratic probing
3. Double Hashing



Topic : Hashing

Linear probing

First probe

probe = h(k)

Zeroth probe

To search Next Location we need offset

key k , h(k)

If location is occupied then

we find the next empty location

probe = h(k)

probe = (probe + 1) % m

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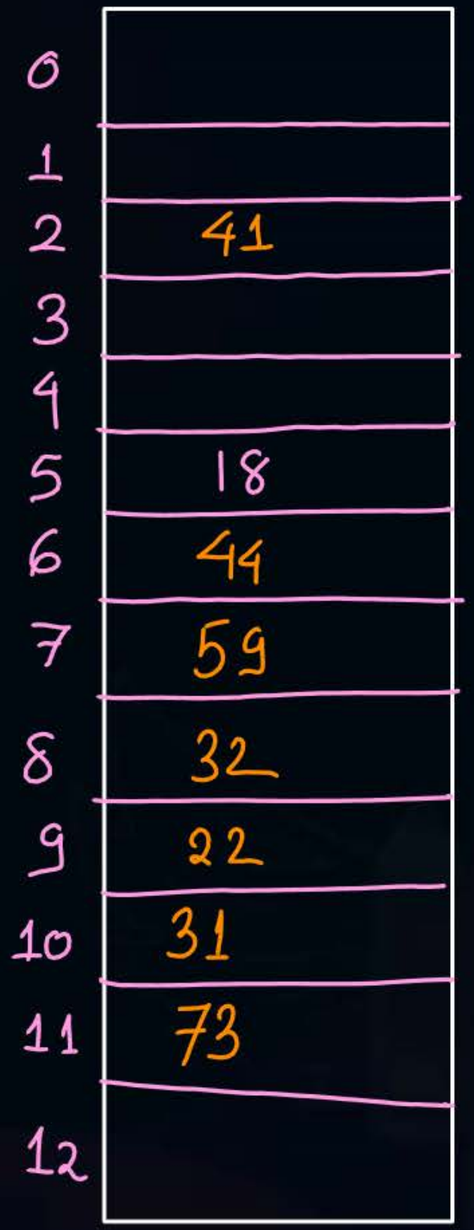
Linear Probing Example

Consider the the hash table of size 13, the hash table is initially empty and hash $h(k) = k \ mod \ 13$ is used. Collision is resolved by *linear probing*.

Insert keys:

18, 41, 22, 44, 59, 32, 31, 73

in to hash table and show the resultant hash table.



① $18\%13 = 5$

$41\%13 = 2$

$22\%13 = 9$

$44\%13 = 5$

$59\%13 = 7$

$32\%13 = 6$

$31\%13 = 5$

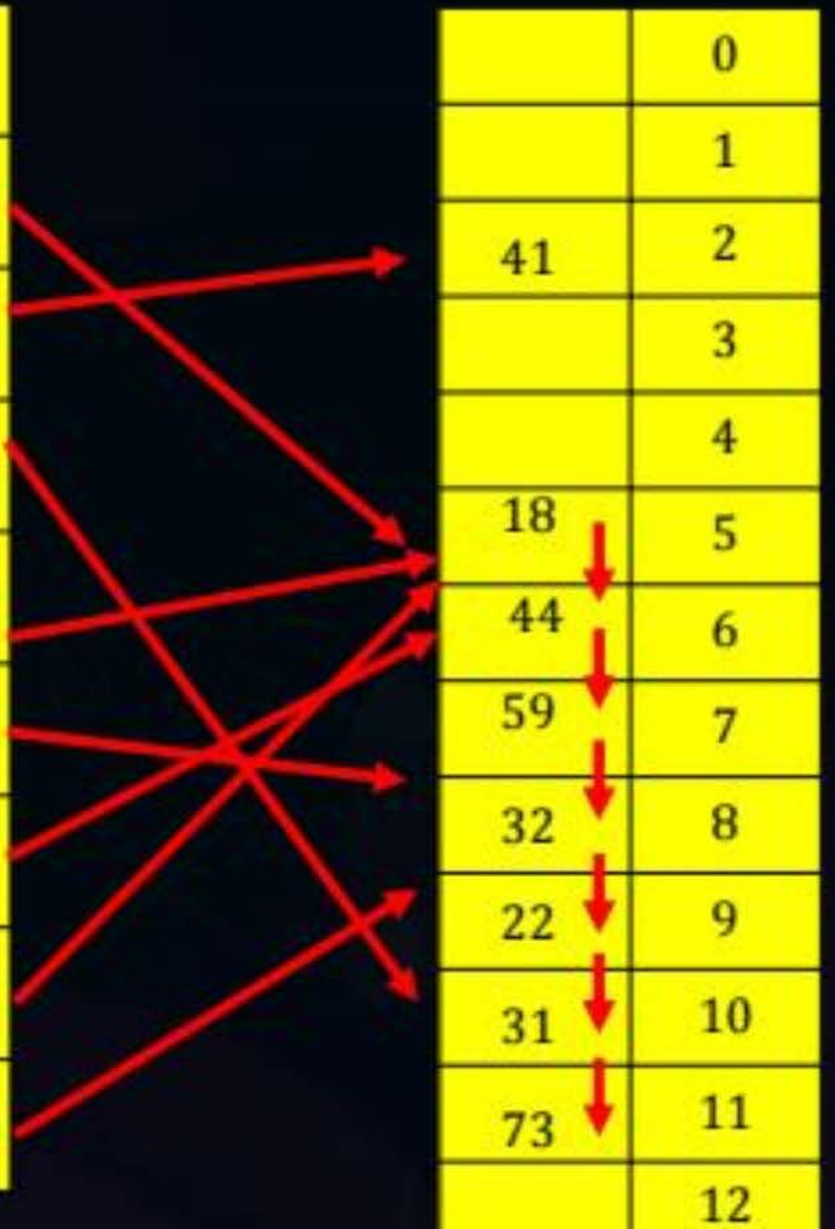
$73\%13 = 8$

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Linear Probing Example

18, 41, 22, 44, 59, 32, 31, 73

Key	$h(x) = x \bmod 13$
18	5
41	2
22	9
44	5
59	7
32	6
31	5
73	8



Slide



Topic : Hashing

A hash table contains 10 buckets and uses linear probing to resolve collisions. The key values are integers and the hash function used is key % 10. if the values 43, 165, 62, 123, 142 are inserted in the table, in what location would the key value 142 be inserted?

(A) 2

(B) 3

(C) 4

(D) 6



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Topic : Hashing

A hash table of length 10 uses open addressing with hash function h(k) = kmod10, and linear probing. After inserting 6 values into an empty hash table, the table is as shown below

0	
1	
2	42
3	52
4	34
5	
6	46
7	33
8	
9	

Which one of the following choices gives a possible order in which the key values could have been inserted in the table?

A) 46, 42, 34, 52, 23, 33

B) 34, 42, 23, 52, 33, 46

C) 46, 34, 42, 23, 52, 33

D) 42, 46, 33, 23, 34, 52

Slide



Topic : Hashing

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Slide



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C) 46, 34, 42, 23, 52, 33

D) 42, 46, 33, 23, 34, 52

Slide



Topic : Hashing

How many different insertion sequences of the key values using the same hash function and linear probing will result in the hash table shown above?

(A) 10

(B) 20

(C) 30

(D) 40

HW

0	
1	
2	42
3	23
4	34
5	52
6	46
7	33
8	
9	

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2 mins Summary

Topic — Direct Addressing

Topic — Hashing

Topic — Chaining

Topic — Linear probing

Topic

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THANK - YOU

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